Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Stallwart

The AI decoy that keeps scammers talking, catches the payee, and turns every scam into protection for the next family.

TEAM
Null & Void

LEADER
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MEMBER
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Theme: BFSI · Intelligent Risk, Fraud and Financial Experiences

BRIEF ABOUT THE IDEA

Catch the scam. Then turn it into the vaccine.

₹22,495 cr

lost to cyber fraud in India in 2025 (NCRP data)

Digital-arrest, KYC and courier scams all end the same way: the victim pays a mule account.

Stallwart answers those calls in English, Hindi and Hinglish, wastes the scammer's time and catches the UPI ID or bank account they want paid.

1 Trap

Ramesh, a worried 68-year-old, keeps the scammer talking and repeating every number.

›

2 Strain

The call becomes a redacted card: the script, the payment method, three warning signs.

›

3 Vaccine

That script becomes a consented practice call for a parent, stopped before any detail leaks.

Why now The RBI's draft mule-account SOP means banks will need evidence leads. Ours are leads, never verdicts.

OPPORTUNITIES

Decoys exist. Protection for the next family does not.

DIFFERENT FROM WHAT EXISTS

Built for India's scripts

- Hindi and Hinglish first
- Digital arrest, KYC, courier, SIM block
- Payee read back and confirmed

Voice decoys exist (Apate.ai). We do not claim to have invented them.

HOW IT SOLVES THE PROBLEM

Time wasted, payee caught

- Every minute on Ramesh is a minute not spent on a victim
- UPI ID or account caught and sent to that bank
- Each script becomes a strain and a drill

OUR USP

Trap, Strain, Vaccine

- Every scam protects the next family
- Two checks before anything is confirmed
- A fresh persona on every call

WHO BUYS IT

Bank fraud teams | UPI apps and fintechs | Telcos | I4C and state cyber cells

LIST OF FEATURES OFFERED BY THE SOLUTION

Everything the loop needs, in one product

1

Live voice decoy

Gemini Live native audio, barge-in, a fresh persona every call

2

English live transcript

Hindi and Hinglish translated on the fly for every viewer

3

Two-layer extraction

Live tool calls plus a Flash verifier. Validators decide

4

Live dashboard

Scam type, trigger phrases, script stage, reply speed

5

Evidence and banks

Recording, transcript and confirmed payees, sent on to the right bank.

6

Strain cards

Redacted script, payment method, three warning signs

7

Family drills

A consented practice call from that day's real script

8

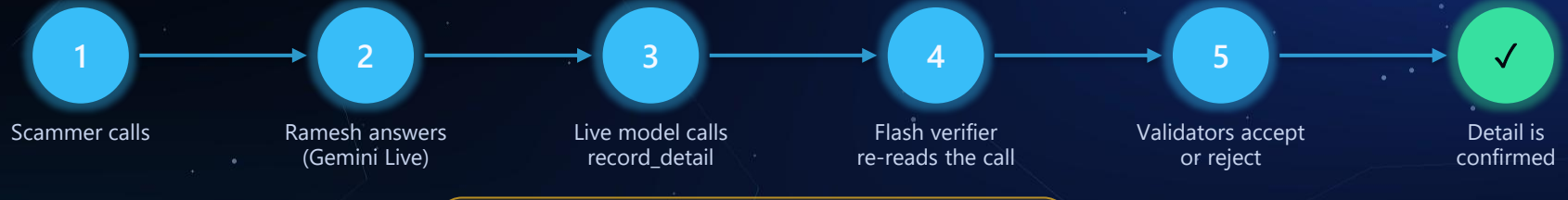
Guardrails and replays

Cost caps, kill switch, two-minute replays with no mic

PROCESS FLOW DIAGRAM

From the first ring to a protected parent

TRAP · ONE SCAM CALL



▼ the finished call feeds the next two stages

STRAIN → VACCINE · THE SAME CALL PROTECTS THE NEXT FAMILY



Two independent checks before anything is confirmed

No identifier survives in a strain

A parent always consents first

NEW · BANK ALERTS

Mule accounts go straight to the right bank

1 **Detail confirmed**
A UPI ID or bank account passes two checks.

2 **Routed to its bank**
The IFSC code, or the UPI handle, names the bank.

3 **Signed alert sent**
Delivered by webhook, retried until it lands.

4 **Bank decides**
Monitor, Freeze or Dismiss. The choice rides on later reports.

₹ **State Bank of India**
Proxy inbox · IFSC prefix SBIN

1 report

BANK ACCOUNT

123456789012

Delivered | Review for freeze | Digital arrest | seen in 2 calls

Play the bank (demo only): Monitor | Freeze | Dismiss

Sample screen, fictional data

One account, many calls
Each account is counted across calls, so repeat mules stand out.

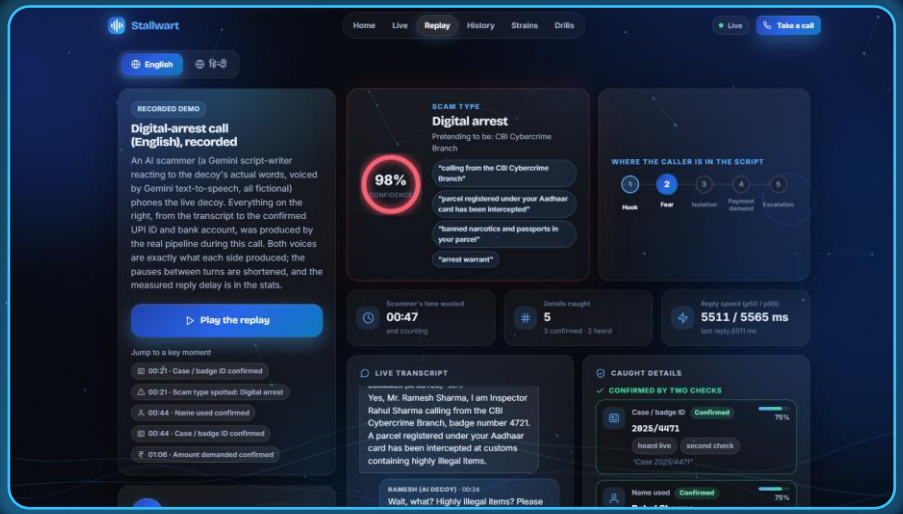
Banks answer back
A freeze or monitor decision rides on every later report.

Proxy banks for now
Nothing is sent to a real bank in this prototype.

WIREFRAMES / MOCK-UPS

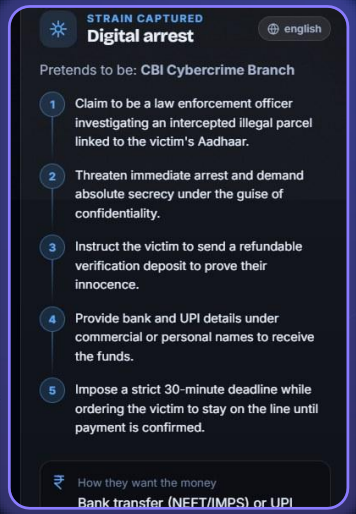
Three screens tell the whole story

Real screens from the live prototype. All names, numbers and IDs are fictional.



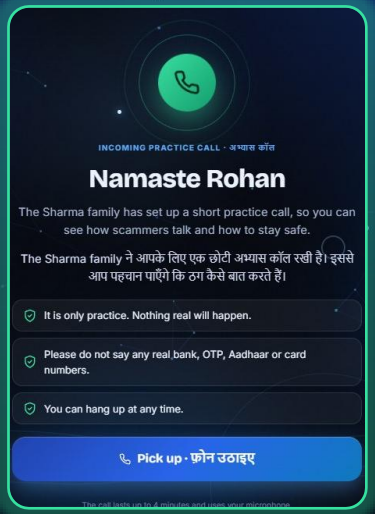
1 Live dashboard

Scam type, trigger phrases, script stage and details that move from heard to confirmed.



2 Strain card

The script with every identifier removed.

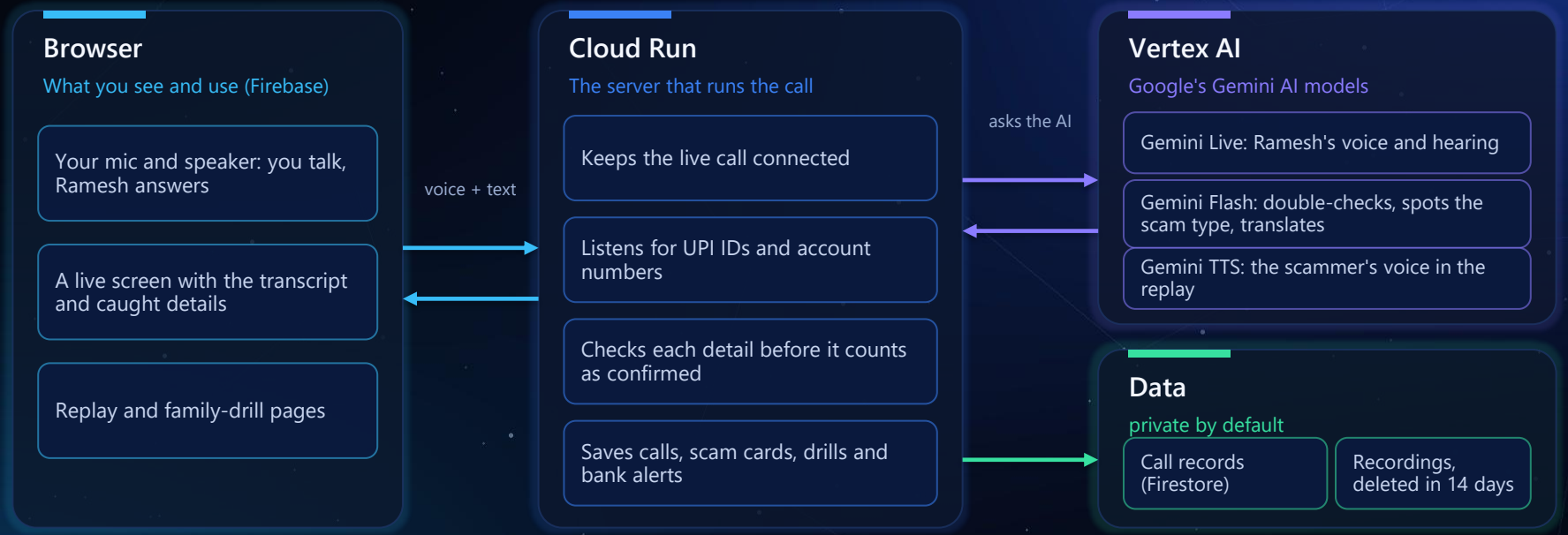


3 Parent's invitation

One tap to a consented practice call.

ARCHITECTURE DIAGRAM

How it works, all on Google Cloud



GUARDRAILS

Per-IP limit | Session cap | 300 s call cap | Max 3 instances | Kill switch

TECHNOLOGIES USED IN THE SOLUTION

A Google-first stack, tested end to end

GOOGLE CLOUD AND AI

- Gemini Live API
- Gemini Flash
- Gemini TTS
- Vertex AI
- Cloud Run
- Firebase Hosting
- Firestore (TTL)
- Cloud Storage
- Cloud Logging
- Billing budgets

APPLICATION

- Python
- FastAPI
- WebSockets
- Pydantic
- ES modules
- Web Audio
- AudioWorklet
- WebGL2 orb
- Strict CSP

QUALITY AND SAFETY

- Deterministic validators
- 114 automated tests
- 42-prompt red-team
- ICFD-31k benchmark
- Decoy variety metrics
- Consent-first drills
- 14-day deletion

Google AI only inside the product. No non-Google model runs in Stallwart.

ESTIMATED IMPLEMENTATION COST

Cents per call, not dollars

$0.01

per two-minute voice call

Measured $0.006 to $0.010 in model tokens, plus a few fractions-of-a-cent Flash requests.

$100

per 10,000 calls a month

Model cost at two minutes each. Cloud Run scales to zero when idle.

$0

to run the prototype

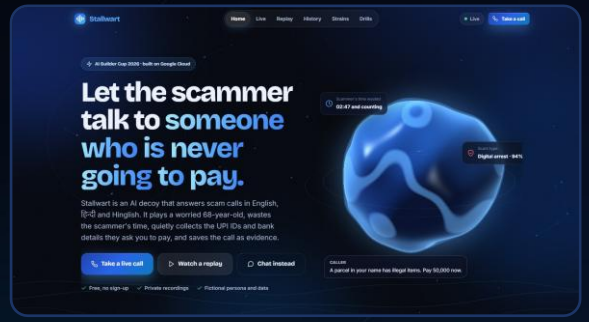
It runs on the free trial credit. Billing is never upgraded.

- Recordings auto-delete after 14 days
- Budget alerts and a one-click kill switch
- Per-IP limits and a 300 s call cap

Estimates from the per-call token estimate the server reports, not invoices. Telephony charges are not included because the prototype runs in a browser.

SNAPSHOTS OF THE PROTOTYPE

The prototype, live today



Home



Live call



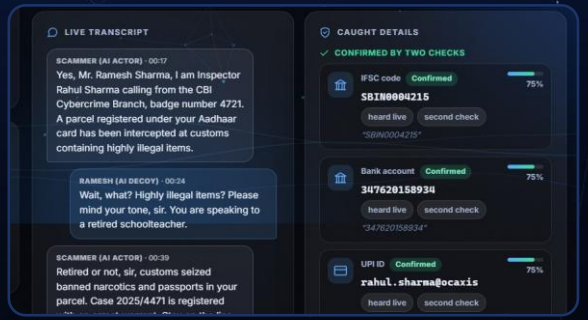
Live dashboard (English call)



Hindi call, English transcript



Call complete: 11 details, 7 confirmed

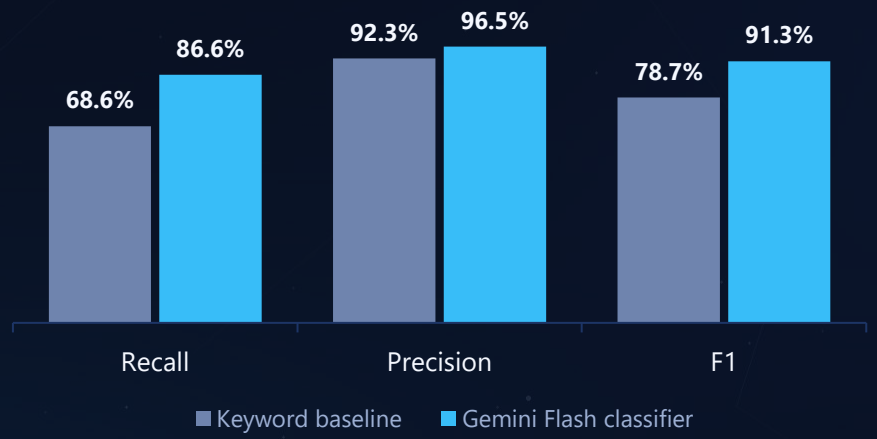


Evidence: confirmed vs heard

PROTOTYPE PERFORMANCE REPORT / BENCHMARKING

Measured, and honest about it

SCAM DETECTION · ICFD-31k · 152 CALLS · 3 RUNS · SYNTHETIC



42 / 42
persona red-team prompts passed (30 hostile + 12 ordinary)

125 / 125
planted spoken-detail tests passed after speech-to-text

37 / 40
fraud calls flagged mid-call (keyword baseline: 31 / 40)

5.5 s
voice-to-voice reply, median (p90 6.2 s)

114
automated tests pass

58% → 0%
decoy replies opening with a filler word

MEDIAN SECONDS INTO A FRAUD CALL BEFORE IT IS FLAGGED



ICFD-31k is synthetic and has only 12 legitimate calls, so specificity is uncertain. The red-team is a single run judged by a second Gemini call.

ADDITIONAL DETAILS / FUTURE DEVELOPMENT

Limits we state plainly, a roadmap we believe in

TODAY'S LIMITS

- A browser stands in for a phone line: no telco diversion yet
- English, Hindi and Hinglish only
- Replies take a few seconds (median 5.5 s)
- Speech can be mis-heard, so every detail is a lead, not proof
- Detection is scored on a synthetic benchmark
- Bank inboxes are proxies: nothing reaches a real bank yet

ROADMAP

1. **Telco SIP diversion**
   Real scam calls reach the decoy
2. **Live bank connections**
   Real bank webhooks and NCRP feeds, with a human check first
3. **Scam radar for banks and I4C**
   Every new strain lights up a shared early-warning map
4. **Voice-clone defence**
   Spot cloned voices and teach families to question them
5. **More Indian languages**
   And faster, more natural voice replies
6. **Vaccine drills on WhatsApp**
   At scale, for families and senior citizens

SAFETY, PRIVACY AND RESPONSIBLE USE

Safe by design, not by promise

Never real data

No OTP, PIN, CVV, card, Aadhaar or PAN. Ever.

Never reveals it is an AI

Passed 42 of 42 red-team prompts. Never insults or threatens.

A lead, not a verdict

Every pack says details came from an unverified caller.

Consent-first drills

A parent agrees first. The call stops at the first sign of risk.

Private by default

Deny-all Firestore, private recordings, deleted after 14 days.

Affordable by design

Per-IP limits, call caps, a kill switch and budget alerts.

LINKS

Try it in thirty seconds

Live product

stallwart.web.app

GitHub repository

github.com/OWAIS285/stallwart

Demo video (3 min)

Link added here after upload

No microphone? Open the Replay in English or Hindi, or type to the decoy in text chat on the Live page.

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Build in JAPAC.
Build for the world.

Thank you

